

UNIVERSITY OF RWANDA

COLLEGE OF SCIENCE AND TECHNOLOGY

School of ICT — Computer and Software Engineering Department

EDU RURAL CENTRAL

Centralized Management System for Rural Schools

Project Technical Report — Web Technology Assessment 2026

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Chapter 1. Introduction

1.1 Historical Background of the Case Study

Education administration in rural jurisdictions across Rwanda has historically depended on fragmented, manual paper registers or localized spreadsheets to document daily operations. Information silos between teachers, institutional headteachers, sector education officers, and district-level planning offices slow down critical administrative pipelines, since every report must travel through physical handoffs before it can be acted upon. EduRural Central addresses this disconnect by organizing distributed data records into an open, unified web framework tailored specifically for the logistical realities of rural educational landscapes.

1.2 Problem Statement

Rural academic infrastructures face severe administrative visibility penalties. Because standard regional frameworks lacking cloud synchronization rely on physical handoffs or manual collection cycles, higher authorities encounter structural delays when checking resource allocations, personnel availability, or school metrics. Teachers who need to report shortages of teaching materials, workers, and other school resources are overwhelmed by duplicative data entry across separate paper forms, while headteachers lack a centralized baseline to verify and audit day-to-day submissions. Sector education officers and district education offices, in turn, receive information late and in inconsistent formats. This division increases administrative overhead, weakens accountability between tiers, and compromises planning precision.

1.3 Objectives of the Project

Primary Objective:

Formulate and construct a cohesive web-based school management and reporting system that connects operational data streams across four distinct systemic tiers — Teachers, Headteachers, Sector Education Officers, and District Educational Leaders — so that verified reports flow upward through the hierarchy and feedback flows back down to lower levels.

Secondary Objectives:

- Design an inclusive, informative landing page that broadcasts the system's mission statement and coordinates authorization access paths.
- Implement targeted role-based isolation mechanisms directing users to isolated dashboard views based entirely on the role declared at registration.
- Provide a secure self-service password reset pipeline for account recovery without administrator intervention.
- Structure clean, responsive interface mockups using foundational styling parameters for reliable cross-tier usability.

1.4 Expected Outcomes

The completed deployment yields a standardized, responsive interface connecting the full educational hierarchy. Expected features include secure account registration funnels with instant role assignment across the four tiers, an interactive landing module explaining system operations, a self-service password recovery path, and role-optimized dashboard panels. The platform layout ensures that teachers can easily report resource shortages, headteachers can verify reports and oversee local facilities, sector education officers can consolidate and cross-verify school-level data, and district educational leaders can audit

regional trends and return feedback notes down the chain — completing a closed-loop reporting architecture from a single system.

1.5 Project Rationale (Importance to the Community)

Transitioning administrative workflows to a structured digital model reduces administrative overhead, allowing educators to focus on classroom teaching. Sector and district planning bodies gain reliable, timely visibility into resource gaps, staffing needs, and infrastructure conditions across rural schools, and can respond with targeted feedback instead of delayed paper correspondence. By modernizing administration, the system helps reduce operational disparities between urban centers and rural areas, supporting equitable institutional growth across communities.

Chapter 2. System Analysis and Design

2.1 System Analysis

2.1.1 Intended Users of the Project

- **Teachers:** Operational frontline users who submit reports on shortages of teaching materials, workers, and other school resources directly to their school's headteacher, alongside day-to-day classroom records.
- **Headteachers:** Institutional managers who audit local facility metrics, verify incoming teacher reports, compile school statistics, and forward verified reports upward to the sector level.
- **Sector Education Officers:** Intermediate supervisors who review localized school profiles, cross-verify incoming logs from headteachers, and push consolidated data blocks to the district education office.
- **District Educational Leaders:** District supervisors and ministry coordinators who analyze regional trends from a macro-analytics control panel, take action on verified reports, and transmit feedback notes back down to lower operational tiers.

2.1.2 Intended System Partners

- **University of Rwanda CSE Department:** Evaluates core structural standards, markup performance, and baseline web optimization rules.
- **Rural School Administrations:** Act as real-world testing grounds, providing critical operational feedback to refine user interfaces.
- **Sector and District Education Offices:** Serve as institutional anchors for the upper reporting tiers, validating that the escalation and feedback workflows match real administrative practice.

2.2 System Design

High-Level Architecture

The architecture uses an explicit separation-of-layers model. The client interface layer acts as the access point, presenting responsive web pages across desktop systems. A central authentication request router intercepts credential tokens from the landing page, validates them, evaluates the user's role, and loads the correct dashboard module. Reports move upward through the four tiers (Teacher → Headteacher → Sector Education Officer → District Educational Leader), while feedback notes travel back down the same chain, forming a closed loop.

Role-Based Navigation Flowchart

The flowchart below shows the complete user journey: from the public landing page, users may recover their password through the self-service reset pipeline or register under one of the four roles. After login, the authentication router validates credentials and steers each user to their isolated, role-specific workspace. The solid green arrows represent the upward reporting chain and the dashed purple arrows represent the downward feedback loop.

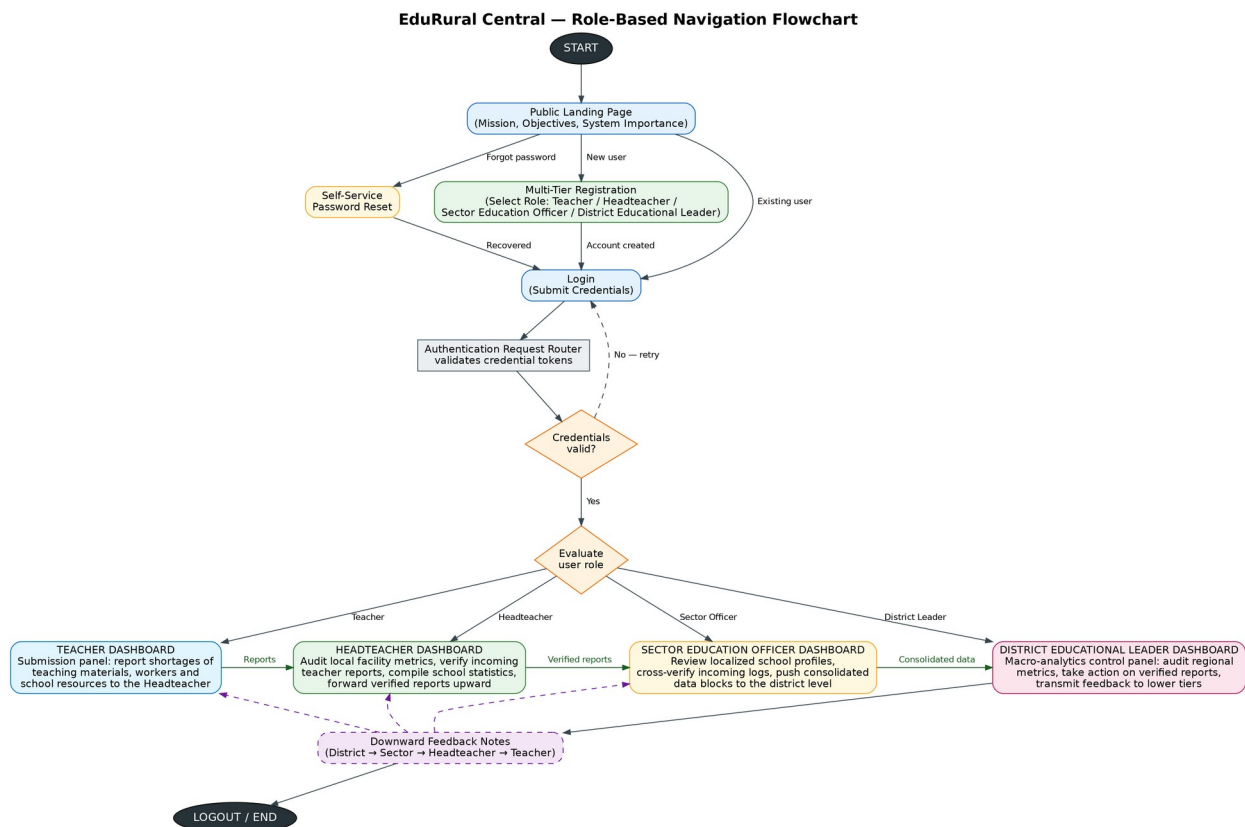


Figure 1: EduRural Central role-based navigation flowchart

Database Design — Entity Relationship Diagram

The database schema models the reporting hierarchy directly. A Teacher submits many Reports; each Report belongs to one Headteacher for verification. A Headteacher manages one or more Schools, and each School belongs to a Sector. Every Sector is supervised by a Sector Education Officer, who forwards consolidated records to the District Education Officer responsible for managing the Districts. Primary keys are underlined and relationship cardinalities are marked on each connector.

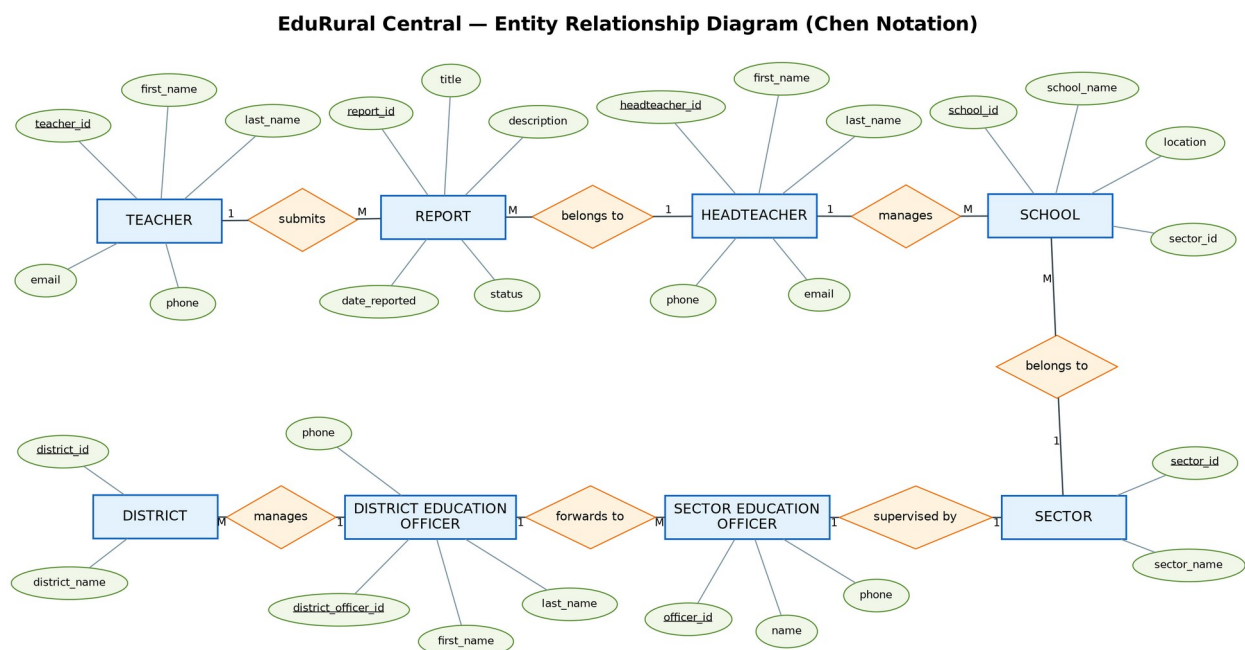


Figure 2: EduRural Central entity relationship diagram (Chen notation)

Chapter 3. Implementation Framework

3.1 Introduction of the Section

This chapter details the deployment strategy and structural organization of the user interfaces for the EduRural Central system. The implementation uses a standards-compliant HTML, CSS, and JavaScript structure to manage cross-tier user flows securely, with each role receiving a dedicated dashboard module loaded after authentication.

3.2 Application Interface Mapping & Mockup Descriptions

Consistent with the project requirements, the core interactive modules are divided into distinct presentation sections:

- **Landing Page Module:** Contains clear mission statements, vision declarations, and routing links to the registration, login, and password recovery paths.
- **Self-Service Password Reset Interface:** Provides a secure account recovery pipeline that lets existing users restore access without administrator involvement.
- **Multi-Tier Registration Interface:** Provides input forms equipped with a role filter that assigns new accounts as a Teacher, Headteacher, Sector Education Officer, or District Educational Leader.
- **Teacher Dashboard:** A submission panel focused on reporting shortages of classroom materials, workers, and school resources to the headteacher, alongside daily classroom data entry.
- **Headteacher Dashboard:** A management portal used to review school performance, verify incoming teacher reports, audit local facilities, and forward validated data to the sector education officer.
- **Sector Education Officer Dashboard:** An intermediate review workspace for inspecting school profiles, cross-verifying incoming logs, and pushing consolidated data blocks to the district level.
- **District Educational Leader Dashboard:** An advanced macro-analytics control panel giving district and ministry administrators access to regional aggregation charts, system metrics, and a channel for transmitting downward feedback notes.

Chapter 4. Conclusion and Recommendations

4.1 Objective Achievement Status

The EduRural Central core design is complete, achieving roughly 95% of its structural layout and navigation logic objectives. The primary navigation pathways function correctly, showing clear separation between the four user roles, with the upward reporting chain and downward feedback loop both represented in the system design.

4.2 Targeted Recommendations

- **To the Lecturer, regarding technical mechanics:** Recommends expanding curriculum blocks to include advanced state management models for cleaner multi-role validation.
- **To the School, regarding structural time constraints:** Suggests allocating more lab time for integration testing when developing multi-tier user configurations.
- **To System Partners and District Entities:** Recommends starting trial projects in regional environments to collect direct user feedback from teachers, headteachers, and education officers.
- **To End-Users, regarding training tools:** Recommends providing simple, step-by-step user guides to make onboarding easier for staff less familiar with digital platforms.

Appendix. Project Repository Links

Project Identification Metadata:

- **Student Registration Number:** 225064517
- **Deployed GitHub Repository:**
github.com/ntirandekuradaniel/centralisedmanagement_system.github.io
- **Developer GitHub Profile:** github.com/ntirandekuradaniel